

Bill of Materials VBF-T2R1SINGLEMODEL-BOM-01

Virtual Battery Factory — t2_r1_singlemodel — 2026-08-25

Component	Mass (g/cell)	Mass (kg/kWh)	Source / formula
Positive active material (NMC811)	22.28	0.89	$I_{\text{pos}} \cdot A \cdot \epsilon_{\text{am}} \cdot \rho = 0.0001 \cdot 0.1027 \cdot 0.665 \cdot 3262$ (parameter set)
Positive conductive additive	Not modeled	Not modeled	Not modeled (Chen2020 has no CB phase)
Positive binder	Not modeled	Not modeled	Not modeled (Chen2020 has no binder phase)
Negative active material (graphite)	31.91	1.275	$I_{\text{neg}} \cdot A \cdot \epsilon_{\text{am}} \cdot \rho = 0.00025 \cdot 0.1027 \cdot 0.75 \cdot 1657$ (parameter set)
Negative conductive additive	Not modeled	Not modeled	Not modeled (Chen2020 has no CB phase)
Negative binder	Not modeled	Not modeled	Not modeled (Chen2020 has no binder phase)
Separator	0.26	0.01	$\text{calc-energy layer_kg_m2} \times \text{area}$
Electrolyte (EC/EMC + LiPF6 + additives)	12.53	0.5	pore volume x 1.2 g/cm3 (literature density, annotated)
Positive current collector (Al)	4.44	0.177	thickness x area x 2700 kg/m3 (parameter set)
Negative current collector (Cu)	11.04	0.441	thickness x area x 8960 kg/m3 (parameter set)
Enclosure / tabs	Not modeled	Not modeled	Not modeled
Total (BOM caliber, incl. electrolyte)	82.19	3.283	calc-energy contract mass excludes electrolyte